CN3104 Cheatsheet AY25/26

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Nonlinear Equations

Goal: Solve $f(x) = 0$.

Graphical Method

-Plot $f(x)$ vs x; roots where curve crosses x-axis.
-Simple but imprecise.

Incremental Search

-Using computer's brute force (relies on trying all possibilities)
1) Divide $[x_{\min}, x_{\max}]$ into n intervals.
2) Find sign changes: if $f(x_i)f(x_{i+1}) < 0$, a root lies inside.
-Use MATLAB function 'incsearch'
-Default number of intervals, ns=50

Disadvantages: -Misses roots if step too large; slow if step too small.

Bracketing Method:Bisection

1) Guess 2 initial starting point, where the root is within. $f(a)f(b) < 0$.(1 positive,1 negative-can apply incremental search/graphical method for initial guess).
2) Iterate midpoint:

$$c = \frac{a+b}{2}.$$

3) If $f(a)f(c) < 0$ set $b = c$, else $a = c$. 4) Stop when stopping criterion met. (threshold error, similar to error in fixed-point iteration)
Guaranteed convergence, slow.

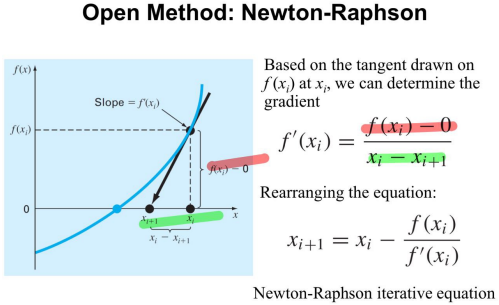
Fixed-Point Iteration

1) Rewrite $x = g(x)$.
2) Iterate $x_{i+1} = g(x_i)$.
3) Calculate error, Error:

$$e_a = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| 100\%.$$

-Stop when error within 0.01 -For bisection, error xi+1 is new root for the present iteration

Newton-Raphson



-Fast convergence.
-Requires derivative and good initial guess.
-Conditions for stop same as secant method.

Secant Method

No derivative needed, uses backward finite difference to estimate f'(x):

$$x_{i+1} = x_i - f(x_i)\frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}.$$

-Requires 2 starting value (do not need to bracket root)
-Conditions for stop: $x_i \approx constant$ or $f(xi) \approx 0$

Systems of Linear Equations

Solve $A\mathbf{x} = \mathbf{b}$.

Gaussian Elimination vs. LU Factorization

Gaussian Elimination (GE): Goal: convert $A \to U$ (upper triangular). Uses row operations:

$$R_2 \leftarrow R_2 - \frac{a_{21}}{a_{11}}R_1.$$

Matrix A is modified during elimination. Does not store multipliers.
LU Factorization: Goal: write $A = LU$. U is obtained from elimination (same as GE). L stores the multipliers used in elimination. To avoid changing A, start L as the identity matrix and insert multipliers.
Identity Matrix Trick:

$$L = I \quad \Rightarrow \quad L(2,1) = m_{21} = \frac{a_{21}}{a_{11}}$$

Insert each multiplier into L while forming U.
Example:

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 5 \end{bmatrix}, \qquad m_{21} = \frac{6}{2} = 3$$

$$U = \begin{bmatrix} 2 & 4 \\ 0 & -7 \end{bmatrix}, \qquad L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$A = LU$$

Forward elimination to make U upper triangular. Then back-substitute.
Pivot equation: eliminate variables below pivot using

$$\text{factor} = \frac{a_{ik}}{a_{kk}}.$$

Ill-Conditioning

Small changes in data cause large changesthe in solution.

Linear Regression

Given data (x_i, y_i).

Simple Linear Regression

Goal: Fit a straight line that best predicts y from x A model is linear if it is linear in the coefficients (parameters). Model:

$$y = a_0 + a_1 x + e.$$

Normal equations:

$$a_0 n + a_1 \sum x_i = \sum y_i$$

$$a_0 \sum x_i + a_1 \sum x_i^2 = \sum x_i y_i.$$

General Least Squares

Matrix form:

$$\hat{y} = Xb$$

$$\varepsilon = y - Xb$$

Minimize $\Phi = \varepsilon^T \varepsilon$.
Solution:

$$b = (X^T X)^{-1} X^T y.$$

Numerical Integration

Goal:

$$I = \int_a^b f(x)\,dx.$$

Trapezoidal Rule

Single interval:

$$I \approx \frac{b-a}{2}[f(a) + f(b)].$$

Composite:

$$I = \frac{h}{2}[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)].$$

Error:

$$E_a = -\frac{(b-a)^3}{12n^2} f''(\xi).$$

Simpson's 1/3 Rule

Requires even number of intervals:

$$I = (b-a)\,\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$$

Composite:

$$I = (b-a)\,\frac{f(x_0) + 4\sum_{\substack{i=1 \\ i \text{ odd}}}^{n-1} f(x_i) + 2\sum_{\substack{j=2 \\ j \text{ even}}}^{n-2} f(x_j) + f(x_n)}{3n}$$

Simpson's 3/8 Rule

$$I = (b-a)\,\frac{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)}{8}$$

Numerical Differentiation

Using finite differences.

Forward Difference

$$f'(x_i) \approx \frac{f(x_{i+1}) - f(x_i)}{h}.$$

Error $O(h)$.

Backward Difference

$$f'(x_i) \approx \frac{f(x_i) - f(x_{i-1})}{h}.$$

Error $O(h)$.

Central Difference

$$f'(x_i) \approx \frac{f(x_{i+1}) - f(x_{i-1})}{2h}.$$

Error $O(h^2)$.

ODEs

Solve:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0.$$

Euler Method

Use finite difference approximation

$$\frac{dv}{dt} \approx \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

Rearranging the expression

$$v_{i+1} = v_i + \frac{dvi}{dt}\Delta t$$

General form:

$$y_{i+1} = y_i + f(t_i, y_i)\, h$$

Heun's Method (Improved Euler)

$$y_1 = y(t_0) + \frac{h}{2}\Big(f(t_0, y_0) + f\big(t_1,\ y_0 + hf(t_0, y_0)\big)\Big)$$

Predictor:

$$k_1 = f(t_i, y_i)$$

Corrector:

$$k_2 = f(t_i + h, y_i + hk_1)$$

Update:

$$y_{i+1} = y_i + \frac{h}{2}(k_1 + k_2).$$

Midpoint Method

$$y_{i+1} = y_i + f\left(t_{i+\frac{1}{2}},\ y_{i+\frac{1}{2}}\right) h$$

$$k_1 = f(t_i, y_i)$$
$$k_m = f(t_i + \tfrac{h}{2}, y_i + \tfrac{h}{2}k_1)$$
$$y_{i+1} = y_i + hk_m.$$

Runge-Kutta 4 (RK4)

$$y_{i+1} = y_i + \frac{1}{6}\big(k_1 + 2k_2 + 2k_3 + k_4\big)h$$

$$k_1 = f(t_i,\ y_i)$$

$$k_2 = f\left(t_i + \frac{h}{2},\ y_i + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_i + \frac{h}{2},\ y_i + \frac{h}{2}k_2\right)$$

$$k_4 = f(t_i + h,\ y_i + hk_3)$$

Adaptive RK (ode45)

MATLAB uses RK4(5) pair. Small step when function changes quickly; large step otherwise.